Scalable Industrial Visual Anomaly Detection With Partial Semantics Aggregation Vision Transformer

Haiming Yao, *Graduate Student Member, IEEE*, Wei Luo, *Student Member, IEEE*, Jianan Lou, *Graduate Student Member, IEEE*, Wenyong Yu, *Senior Member, IEEE*, Xiaotian Zhang, Zhenfeng Qiang, and Hui Shi, *Member, IEEE*

***Abstract*— In recent years, the field of industrial visual anomaly detection (VAD) has attracted significant attention in the context of advanced smart manufacturing systems. However, several limitations remain unresolved in existing approaches. While these methods can achieve satisfactory performance when training separate models for different categories, their scalability and performance suffer when faced with the challenge of simultaneous training for multiple categories. Reconstruction-based methods generally suffer from the identical mapping problem. To address these limitations, this study introduces the partial semantic aggregation vision transformer (PSA-VT), a scalable framework for industrial visual anomaly detection (VAD) that enables simultaneous multicategory anomaly detection using a single model. Our proposed PSA-VT framework adopts a hybrid design strategy. First, a pretrained convolutional neural network (CNN) is employed to extract multiscale discriminative local representation. Subsequently, the PSA-VT is introduced to perform representation reconstruction through long-range global semantic aggregation. Finally, the anomalous properties can be estimated by evaluating the reconstruction error of the representations. We conducted extensive experiments using the Mvtec AD industrial anomaly detection dataset, as well as the semantic anomaly detection datasets. The experimental results demonstrate that our method achieves state-of-the-art (SOTA) performance by capturing high-level semantics. Notably, PSA-VT surpasses other methods for the one-model-15-category anomaly detection tasks on the Mvtec AD dataset. Furthermore, we applied incremental learning techniques to enable the rapid deployment of PSA-VT in a real industrial scenario.**

***Index Terms*— Anomaly localization, defect inspection, scalable visual anomaly detection (VAD), self-supervised learning, unsupervised learning, vision transformer (ViT).**

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Haiming Yao, Wei Luo, Jianan Lou, Xiaotian Zhang, and Zhenfeng Qiang are with the State Key Laboratory of Precision Measurement Technology and Instruments, Department of Precision Instrument, Tsinghua University, Beijing 100084, China (e-mail: yhm22@mails.tsinghua.edu.cn; luow23@mails.tsinghua.edu.cn; ljn22@mails.tsinghua.edu.cn; zhangxt6@foxmail.com; 18302973462@163.com).

Wenyong Yu is with the State Key Laboratory of Digital Manufacturing Equipment and Technology, School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: ywy@hust.edu.cn).

Hui Shi is with the School of Mechanical and Electronic Engineering, Wuhan University of Technology, Wuhan 430070, China (e-mail: clove_shi@whut.edu.cn).

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I. Introduction

AUTOMATIC visual anomaly detection (VAD) has garnered considerable attention and widespread implementation across various sectors in the smart industry, due to its highly automated, nondestructive, and efficient detection capabilities. VAD has found practical applications in a range of industrial scenarios, including product quality inspection [1], textured surface defect detection [2], [3], and smart power systems [4], among others. The utilization of VAD offers substantial improvements in both efficiency and accuracy compared with traditional manual inspection methods.

Supervised learning-based methods and unsupervised learning-based methods represent the two predominant approaches in industrial VAD. Supervised methods, leveraging annotated datasets consisting of numerous defective samples and their corresponding labels, have demonstrated robust detection performance [4], [5] across diverse scenarios. Nevertheless, the process of collecting and annotating a large number of defect samples is often impractical and time consuming in real-world industrial settings. Consequently, Wan et al. [1], Yao et al. [2], and Yang et al. [3] have turned their attention to unsupervised methods, which possess the advantage of utilizing only normal nondefective samples without the need for any annotations.

The field of unsupervised VAD commonly adopts a method of learning the distribution of normal patterns through reconstruction. The underlying assumption is that only normal samples are utilized during the model training, and the objective is to generate reconstructions that closely resemble the input. During testing, both normal and abnormal samples will result in reconstructions that resemble the normal distribution. The discrepancy between the input and the reconstruction is then employed as the criterion for anomaly detection. This approach has been extensively studied in subsequent research endeavors. However, prevalent reconstruction-based VAD methods suffer from a significant limitation known as “trivial solutions” or “identical mapping.” This refers to instances where the model simply outputs a direct replica of the input during the reconstruction process, without considering its semantics. As a consequence, even outlier samples can obtain high-fidelity reconstructions, posing a challenge for anomaly discrimination in reconstruction-based methods.

In practical VAD, the detection of anomalies across different categories remains an underexplored area. This issue holds significant relevance in real-world industrial environments, as illustrated in Fig. 1. Traditional methods encounter challenges when multiple product types require simultaneous detection, necessitating the utilization of independent model parameters for each product category. Consequently, this leads to substantial memory consumption for storage and inference purposes. To tackle this problem, we propose a scalable model that offers a comprehensive solution adaptable to inspections of all product types through the use of a single set of unified parameters. Unlike traditional methods that become increasingly impractical as the number of product types expands due to the need for separate parameter storage for each category, our proposed scalable approach requires the storage of only one set of generic parameters. This characteristic allows for scalability and versatility, regardless of the number of product types involved.

The innovative multicategory anomaly detection approach poses a greater challenge than the conventional single-category method. The model is required to learn the normal distribution of multiple categories and perform the reconstruction process for each distinct category simultaneously, which may lead the model to further favor "trivial solutions" without considering semantic information. Moreover, the inference process requires the model to detect multiple categories without explicit label information or fine tuning, and thus, the model should not exhibit any biases toward a specific category. Such biases may result in the misclassification of samples from other categories as anomalous, even if they are normal or anomalous. This highlights the crucial importance of the model's ability to comprehend the global semantics of objects.

CNNs have been widely used and achieved impressive performance in various vision tasks, including VAD. However, their limited receptive field hinders them from comprehending global semantics, leading to reduced discriminability between normal and defective patterns [6]. To address this issue, the vision transformer (ViT) was proposed [7], which uses the self-attention mechanism to model images from a sequence perspective and capture global context. This promising methodology has the potential to address the inherent shortcomings of CNNs in the context of VAD and has garnered substantial attention in recent research endeavors.

Drawing from the preceding analysis, we have devised a novel hybrid reconstruction framework called partial semantic aggregation vision transformer (PSA-VT) to capitalize on the strengths of both convolutional neural networks (CNNs) and ViTs, enabling a scalable approach to cross-category VAD in industrial scenarios. In the PSA-VT framework, CNNs are first employed to extract multiscale discriminative local representation, effectively transforming image pixels into deep feature maps imbued with local semantics. Subsequently, the partial semantic aggregation paradigm is introduced within ViT, incorporating auxiliary aggregation tokens to extract the global semantic context of the original feature's patch tokens. In this paradigm, the reconstruction task transforms into a self-supervised semantic aggregation process. This study offers two primary contributions.

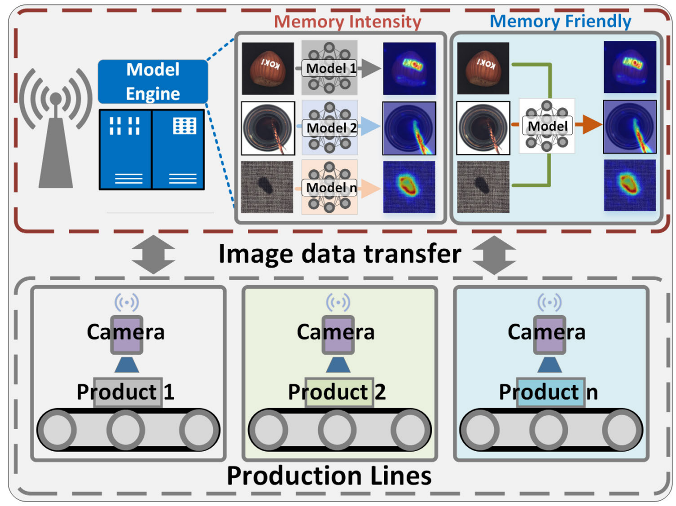

Fig. 1. Real-world industrial settings demand simultaneous testing of numerous products. Conventionally, the inspection of captured image data from diverse product lines involves the use of multiple models in memory-intensive methodologies. However, the proposed innovative generalizable model provides a compelling alternative, requiring only a single model to effectively detect multiple products simultaneously.

1) We envisioned a challenging yet practical detection framework for industrial applications: scalable multi-category anomaly detection. To achieve this objective, we have proposed a novel hybrid VAD approach, called the PSA-VT. The PSA-VT framework comprises two key modules: a local feature representation module, built upon a pretrained CNN and a global feature reconstruction module, powered by the ViT within the partial semantic aggregation paradigm. Specifically, the feature extraction module extracts the feature maps with local contexts as descriptors. On the other hand, the feature reconstruction module utilizes a global context aggregation paradigm to reconstruct abnormal features into normal patterns.
2) To assess the efficacy of our proposed PSA-VT method, a comprehensive evaluation is conducted on the widely utilized MVTec AD dataset. The evaluation results demonstrate that PSA-VT achieves SOTA performance in terms of AUROC, with 96.4 for anomaly detection and 96.9 for anomaly segmentation, in the context of scalable multicategory anomaly detection (one-model-15-category setting). To gain a deeper understanding of the underlying mechanisms and factors influencing performance, ablation experiments are performed along with related analyses. Moreover, in order to ascertain the practical applicability of our method in real-world industrial scenarios, incremental learning techniques are employed, and the effectiveness of our method is validated.

The rest of this article is organized as follows. Section II offers a literature review that encompasses pertinent studies on VAD, with particular emphasis on reconstruction-based and ViT-based methods. Section III elaborates on the intricacies of our proposed approach, the PSA-VT. Section IV presents a thorough analysis of the experimental findings. Finally, Section V summarizes the conclusion and provides a concise summary of this article.

II. Related Works

A. Reconstruction-Based Anomaly Detection

The extensive research has been undertaken in the domain of VAD, encompassing various methodologies. Notably, embedding-based approaches [8], [9] have been explored, which involve the modeling of normal feature distributions in a latent space. These approaches utilize the dissimilarity between instances and normal feature clusters as a discriminative metric. However, for the purpose of this study, our focus is specifically directed toward reconstruction-based approaches in VAD.

Deep autoencoders (AEs) are the most popular model in reconstruction-based VAD methods. To address the limitations of standard AE, several extensions have been proposed to enhance their capabilities in anomaly detection. For instance, the memory mechanism was proposed in the MemAE model [10] that explicitly stores the prototypical normality patterns to restrict the high-fidelity anomaly reconstruction issue encountered in vanilla AE. Similarly, the partition memory technique proposed in [11] ensures the preservation of normal semantic integrity. In addition, the multilevel image reconstruction (MLIR) model [12] generates multilevel reconstruction outcomes with different retentions, while the reconstruction by inpainting for anomaly detection (RIAD) model [13], augmented with attention in [14], leverages image inpainting techniques to improve the reconstruction process. Similar concepts are also explored in [15], where predictive regression is utilized within the local receptive field of CNNs as an alternative to direct reconstruction. In addition, the novel Draem method introduced in [16] introduces an extra pseudo-anomaly and discriminative subnetwork to enhance anomaly segmentation. While most existing methods focus on the reconstruction difference at the pixel level, the deep feature reconstruction (DFR) approach [17] employs feature reconstruction difference. DFR leverages the local semantic perception capability of the CNN network, enabling it to extract multilevel features and perform feature-level reconstruction, thereby achieving more robust anomaly detection performance. To address the issue of over generalization in feature reconstruction, the dual-Siamese network [18] incorporates synthetic defects and introduces feature repair techniques to enhance the stability of the reconstruction process. Moreover, approaches such as [1] and [6] employ feature mapping and correspondence mechanisms to establish diverse normality relations across different feature domains, thereby expanding the applicability of reconstruction-based methods. However, it is noteworthy that most existing methods, including the aforementioned ones, are predominantly based on the CNN model. The local semantic awareness of CNNs may limit the effectiveness of reconstruction-based methods when dealing with challenging defect types that exhibit global semantics. Moreover, the aforementioned approaches are developed within the framework of separate training for each category and do not investigate the one-model-multicategory scalability [19], which limits their applicability in scenarios, where multiple categories of production need to be inspected simultaneously.

B. Transformer Model in Anomaly Detection

The ViT, a novel architecture in the field of computer vision, has demonstrated remarkable efficacy across various visual tasks [20]. However, its utilization in the VAD community, which predominantly relies on CNN-based methods, has been relatively limited. To address this gap, several models have been proposed that incorporate ViTs into the VAD framework. The inpainting transformer (InTra) was introduced in [21], offering a solution for image reconstruction through training a model to inpaint the missing patch by the surrounding patches. In a similar vein, U-transformer [22] tackles feature reconstruction by employing a transformer inspired by the U-net structure. Furthermore, MSTUnet [23] adopts a strategy that combines defect synthesis and image inpainting and incorporates the Swim transformer to leverage its long-range modeling capability for inpainting purposes.

This study presents a novel reconstruction method termed PSA-VT. By leveraging the long-distance semantic dependency of the ViT, PSA-VT adopts a paradigm of partial semantic aggregation to accomplish self-supervised feature reconstruction. This approach overcomes the drawbacks of identity mapping observed in existing reconstruction-based methods, establishing a new paradigm for scalable multiclass VAD.

III. Proposed Methodology: PSA-VT

A. Theoretical Analysis and Motivation

In order to investigate the occurrence of the "identical mapping" phenomenon during the reconstruction process, we begin with a theoretical inquiry into the reconstruction task, focusing on the underlying mechanisms of various neural networks like [19]. In the case of CNNs, the output $\mathcal{Y}$ is a function of the input $\mathcal{X}$, determined by a set of weight coefficients ($\mathbf{W}$) and a bias term ($\mathbf{B}$) and can be represented by the following equation:

$$\mathcal{Y} = \mathbf{W} \cdot \mathcal{X} + \mathbf{B}. \tag{1}$$

To achieve the reconstruction of the input $\mathcal{X}$ as $\mathcal{Y} = \mathcal{X}$, a CNN can assign the weight coefficient $\mathbf{W}$ as the identity matrix $\mathbf{I}$ and the bias $\mathbf{B}$ as 0. However, this approach results in a "trivial solution," where the network fails to capture the input's true semantic nature. On the other hand, the mechanism underlying the self-attention-based ViT can be summarized as follows:

$$\mathcal{Y} = \underbrace{\left(\xi\left(\theta_q(\mathcal{X}) \cdot \theta_k(\mathcal{X})\right)^{\mathsf{T}}\right)}_{\mathbf{A}} \theta_v(\mathcal{X}) + \mathcal{X} \tag{2}$$

where θ is the linear transformation, ξ is the soft-max function, and $\mathbf{A}$ represents the attention matrix. In order to reconstruct the input and achieve $\mathcal{Y} = \mathcal{X}$, the model may learn to set θ_v and attention matrix to $\mathbf{I}$. However, this also leads to the "trivial solution," where the attention mechanism of each token is more focused on its own information, rendering the global attention feature of the transformer model ineffective. As a result, the global semantics of the object cannot be learned. In addition, residual connections that may lead the information leakage within the model can also be an important shortcut for the reconstruction task.

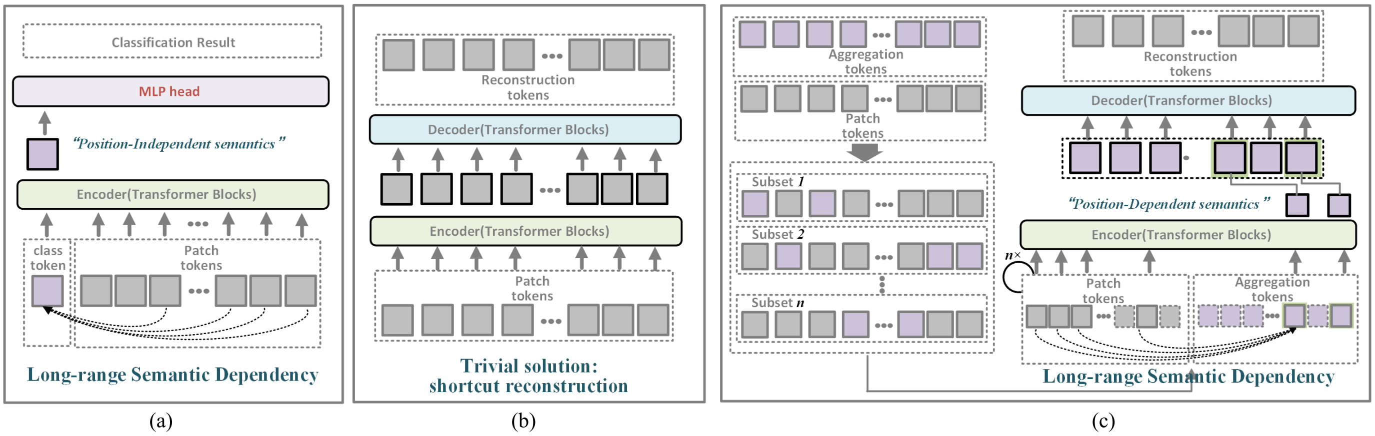

Fig. 2. (a) ViT [7] employs a class token to capture long-range semantic dependencies among the original image's patch tokens for visual classification. (b) Directly using the patch tokens for the reconstruction task with the vanilla transformer risks falling into the trivial solution of shortcut reconstruction. (c) Our proposed PSA-VT introduces aggregation tokens to aggregate global semantic representations while preserving spatial information by aggregating the information of position-disjoint patch tokens under the supervision of the reconstruction task.

To effectively tackle the challenges of multicategory anomaly detection tasks and avoid the aforementioned trivial solutions, it is crucial for a model to have a comprehensive grasp of object semantics. Drawing from the approach adopted by ViT [7], which employs a class token for global semantic aggregation in visual classification, we propose a novel strategy that leverages auxiliary aggregation tokens. As shown in Fig. 2(a), ViT introduces an additional class token to accomplish visual classification tasks by aggregating high-level global semantic information. However, as depicted in Fig. 2(b), when employing the standard transformer model for the reconstruction task, the "trivial solution" mentioned earlier may occur, especially when attempting to match intricate multicategory distributions.

To address the above problems, we present a novel partial semantics aggregation transformer with auxiliary aggregation tokens that enables the model to extract high-level global semantics while preserving spatial information for the reconstruction task. As shown in Fig. 2(c), our novel approach goes beyond ViT [7] by generating representations with holistic global semantics for all spatial locations in the self-supervised learning paradigm. Specifically, we utilize a set of learnable aggregation tokens that share the same shape as the original patch tokens. This ensures that each position in the original feature sequence has an aggregation token. The aggregation tokens are divided into subsets that are associated with feature patch tokens from disjoint positions, facilitating the learning of long-range semantic dependencies during the reconstruction task. As a result, our method can obtain position-dependent global semantics for each position. In Section III-B we shall elaborate on this approach in a comprehensive manner.

B. Overall Pipeline

Fig. 3 illustrates the overarching architecture of our proposed PSA-VT. Serving as a hybrid framework, it combines the strengths of both CNN and ViTs, thereby encompassing local and global semantic awareness. The framework is primarily composed of two modules: the local feature representation module and the global feature reconstruction module. Initially, the input image undergoes conversion into a multiscale discriminative local representation via a pretrained CNN. Subsequently, the PSA-VT is employed to reconstruct the extracted representation by leveraging global contextual semantic aggregation.

In the proposed partial semantic aggregation paradigm, we initially transform the feature representation into a sequence of feature patch tokens. To facilitate the aggregation process, we introduce an auxiliary aggregation label sequence that shares the same shape as the feature patch token sequence. The aggregation token sequence is partitioned randomly into multiple disjoint subsets. Each subset is then combined with a set of partial feature patch tokens whose positions do not overlap, forming a new set of sequences. These sequences are subsequently input to the transformer encoder.

The aggregation tokens within each sequence enable the perception of information from feature patch tokens that occupy nonintersecting positions. This allows for the aggregation of feature patch tokens through long-distance semantic dependencies, resulting in the global semantics of the partial feature patch tokens for each sequence being aggregated in aggregation tokens after encoding. In the latent space, all aggregation tokens from across all sequences are merged, while the raw feature patch tokens are discarded. The resulting a new combined sequence, which is then processed by the decoder to reconstruct the original features.

In Sections III-C–III-G, we will provide detailed explanations of each module of the PSA-VT method.

C. Local Feature Representation With Pretrained CNN

To derive local discriminative semantic representations from input images $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$, we employ a process that involves converting the raw pixel values into CNN deep feature maps. Consistent with the previous research [17], we utilize a pretrained CNN that has been trained on visual recognition tasks as our feature extractor. This allows us to leverage CNN's capability to extract intricate local structures, thereby obtaining

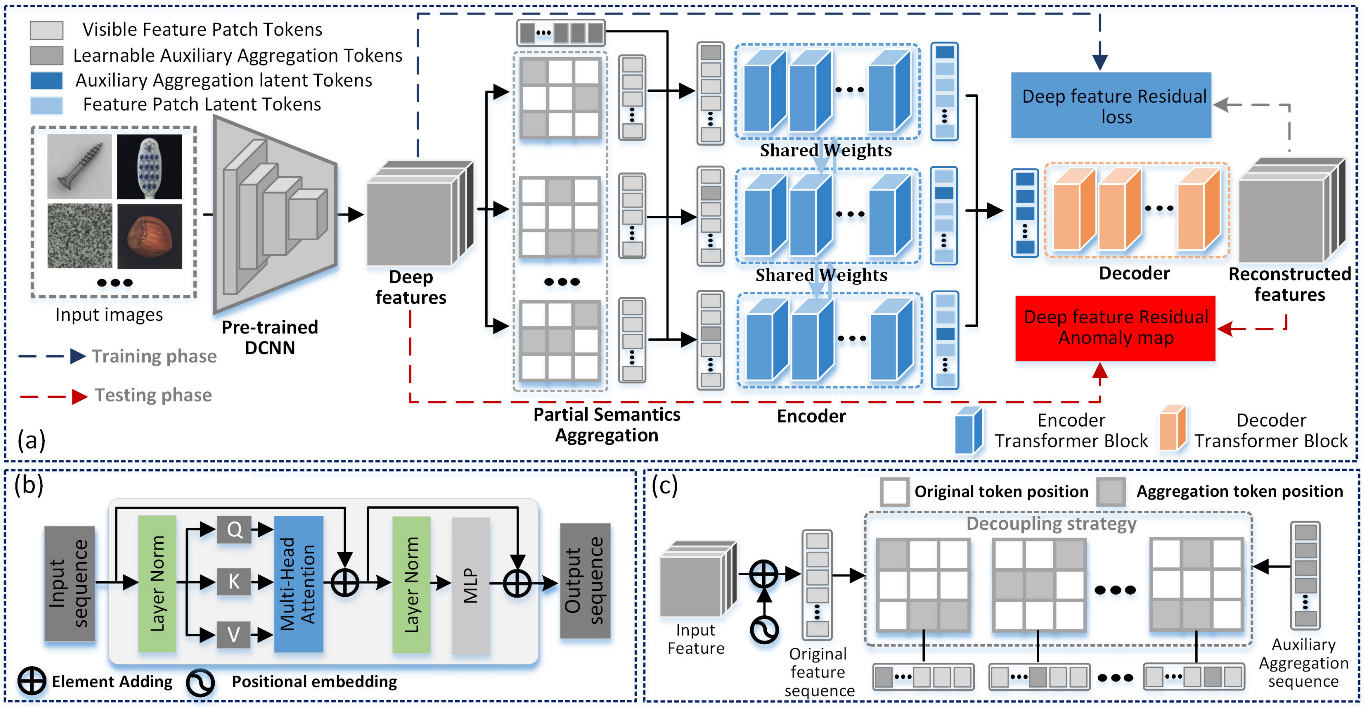

Fig. 3. (a) Overall pipeline of the proposed PSA-VT. First, the local feature representation module employs a pretrained CNN to extract multiscale discriminative local representations for the input image. Subsequently, the global feature reconstruction module utilizes the partial semantic aggregation paradigm, facilitated by a ViT, to accomplish representation reconstruction. (b) Typical computational process of self-attention in the basic ViT block, as outlined in [7], is as follows. (c) Schematic illustrates the partial semantic aggregation paradigm.

local representations that possess discriminative local semantic information.

Unsupervised VAD methods lack prior knowledge about defects, including their sizes. To capture local structural representations across various scales, we employ a multiscale feature extraction strategy. Specifically, different levels of CNN possess feature maps with distinct receptive field sizes. Hence, we selectively extract features from multiple CNN levels. To enable the fusion of these multiscale features, we first adjust them to a consistent spatial size through interpolation. Subsequently, we concatenate these features along the channel dimension. This process yields the multiscale discriminative local representation $\mathcal{F} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$.

By utilizing deep feature maps instead of low-level pixel representations, we exploit the advantages of capturing receptive information with local semantics. $\mathcal{F}$ provides a more locally discriminative representation of the input image, which enhances our ability to reveal and detect anomalous patterns. This approach allows us to leverage the power of the pretrained CNN's learned features to improve the accuracy and robustness of our anomaly detection framework.

Algorithm 1 Reconstruction by PSA

Input: Input feature $\mathcal{F}$
Hyper-parameters: Patch size P; Subset number N
Result: Reconstructed feature $\tilde{\mathcal{F}}$

Step1.Embed the $\mathcal{F}$ into sequence: $\mathcal{X} = \mathbf{E}(\mathcal{F}, P)$;
Step2.Given the auxiliary aggregation sequence $\mathcal{X}^*$;
Step3.Randomly divide $\mathcal{X}^*$ into N disjoint token subsets:
$\left\{\mathcal{X}_i^*, i = \{1, \ldots, N\} \mid \cup_i^N \mathcal{X}_i^* = \mathcal{X}^*\right\} = D(\mathcal{X}^*, N)$;
Step4.**for** $i = 1, \ldots, N$ **do**
  Replace the corresponding tokens in $\mathcal{X}$ with $\mathcal{X}_i^*$: $\Phi_i = \mathcal{X}_i^* \cup \complement_{\mathcal{X}} \mathcal{X}_i^*$;
end
Step5.Encode the $\mathbf{\Phi} = \{\Phi_i, i = \{1, \ldots, N\}\}$ with **Enc**:$\mathbf{Z}_{\mathbf{\Phi}} = \{Z_{\Phi_1}, Z_{\Phi_2}, \ldots, Z_{\Phi_N}\} = \mathbf{Enc}(\mathbf{\Phi})$;
Step6.**for** $i = 1, \ldots, N$ **do**
  Retain the aggregation tokens embedding in Z_{Φ_i}: $Z_{\mathcal{X}^*} = Z_{\mathcal{X}^*} \cup Z_{\Phi_i}(\mathcal{X}_i^*)$;
end
Step7.Decode the $Z_{\mathcal{X}^*}$ with **Dec**: $\tilde{\mathcal{F}} = \mathbf{Dec}(Z_{\mathcal{X}^*})$;
return $\tilde{\mathcal{F}}$

D. Global Feature Reconstruction With PSA-VT

In order to address the requirement for global semantic awareness in scalable multicategory VAD, we propose the partial semantics aggregation paradigm driven by ViTs. This paradigm aims to reconstruct the extracted representation $\mathcal{F}$ from a global semantic perspective. The reconstruction process employing partial semantic aggregation is succinctly outlined in Algorithm 1, accompanied by a visual representation of each sequential step illustrated in Fig. 4(a).

First, to facilitate the processing of the 2-D feature maps, we initially reshape $\mathcal{F} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$ along the spatial dimension. This reshaping operation transforms the feature maps into a sequence of flattened 2-D patches, each having a size

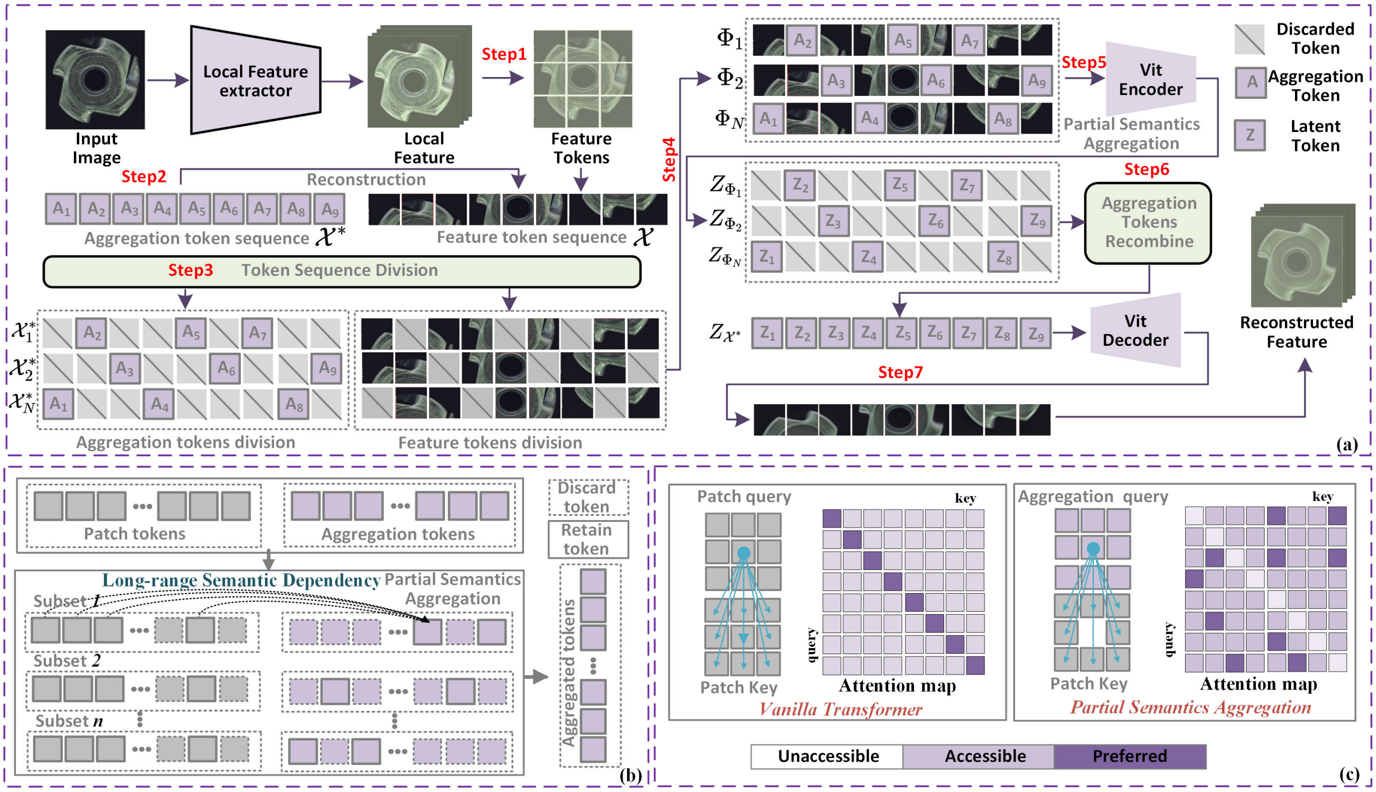

Fig. 4. (a) Working principle of the PSA-VT. Each step in Algorithm 1 is shown in this figure, respectively. (b) Schematic of the partial semantic aggregation process, where auxiliary aggregation tokens in each subset establish long-distance semantic dependencies with its disjoint patch tokens. (c) Comparison of the self-attention mechanism between the vanilla transformer and the proposed partial semantic aggregation transformer, the vanilla transformer has a tendency to focus on its local information when performing the reconstruction task, while the partial semantics aggregation transformer establishes long-range global semantic dependencies.

of $P \times P \times \mathcal{C}$, which are then mapped into D dimensional embeddings. This division strategy enables the construction of a feature patch sequence $\mathcal{X}$ with a shape of $(\mathcal{H}\mathcal{W})/P^2 \times D$. By employing this patch-level division approach, we ensure a more efficient computational process.

In this article, we present a novel self-supervised approach based on ViTs for the global reconstruction task of feature patch token sequence $\mathcal{X}$. As elaborated in Section III-A, the direct utilization of feature patch token sequences for self-reconstruction often leads to the "identical mapping problem." To alleviate this challenge, as illustrated in Fig. 4(a), we introduce an auxiliary aggregation token sequence denoted by $\mathcal{X}^*$ to facilitate the reconstruction of the feature patch token sequence $\mathcal{X}$. To enable effective reconstruction of $\mathcal{X}$, the auxiliary aggregation token sequence $\mathcal{X}^*$ is meticulously designed to match the shape of the feature patch token sequence $\mathcal{X}$, ensuring that each feature patch token corresponds to a counterpart auxiliary aggregation token. To tackle this reconstruction task, we introduce a novel technique termed "partial semantic aggregation" for achieving self-supervised global reconstruction of the feature patch token sequence.

The operational framework of partial semantic aggregation is systematically illustrated in Fig. 4(a), delineating each step of the process. To incorporate long-range semantic dependencies effectively, in step 3, we employ a random token sequence division strategy for $\mathcal{X}^*$, breaking it down into N distinct subsets denoted by $(\mathcal{X}_1, \mathcal{X}_2^*, \ldots, \mathcal{X}_N^*)$, where each subset consists of $(1/N)$ aggregation tokens. Conversely, for $\mathcal{X}$, a complementary partitioning approach is adopted. Specifically, this involves ensuring that the tokens within each subset of $\mathcal{X}$ and the tokens within the corresponding subset of $\mathcal{X}^*$ exhibit a complementary positional distribution. Subsequently, in step 4, we merge N corresponding subsets that were partitioned from $\mathcal{X}^*$ and $\mathcal{X}$, giving rise to the generation of N new sequences

$$\boldsymbol{\Phi} = \left\{\Phi_1, \Phi_2, \ldots, \Phi_N | \Phi_i = \mathcal{X}_i^* \cup \complement_{\mathcal{X}} \mathcal{X}_i^*, i = 1, \ldots, N\right\} \tag{3}$$

where $\complement_{\mathcal{X}} \mathcal{X}i^*$ represents the feature patch tokens within each sequence that do not overlap with the aggregation tokens from the positioning perspective. Thus, for these N sequences, each sequence Φ_i consists of aggregation tokens comprising $(1/N)$ of the total tokens, while the remaining $(N-1)/N$ proportion is comprised of disjoint feature patch tokens.

In step 5, we feed this collection of sequences $\boldsymbol{\Phi}$ into the transformer encoder. This process yields a latent space representation for each sequence within the set: $\mathbf{Z}_{\boldsymbol{\Phi}} = \{Z_{\Phi_1}, Z_{\Phi_2}, \ldots, Z_{\Phi_N}\}$. During the encoding phase, the transformer effectively aggregates partial semantic information from the feature patch tokens into the aggregation tokens, leveraging long-range semantic dependencies, as illustrated in Fig. 4(b). The aggregation tokens within each sequence capture global contextual normality information from their respective nonoverlapping feature patch tokens. Moving to the latent space in step 6, we eliminate the feature tokens within each sequence, as they may potentially contain abnormal information. Instead, we only extract the aggregation tokens from each sequence and recombine them to form a complete sequence, denoted by $Z_{\mathcal{X}^*}$. Subsequently, in step 7, we pass $Z_{\mathcal{X}^*}$ through the decoder to execute the reconstruction task on the feature patch sequence. Finally, the resulting sequence is remapped to the 2-D space, yielding the reconstructed feature $\tilde{\mathcal{F}}$.

Fig. 4(c) presents a comparison between traditional transformers and our proposed partial semantic aggregation transformers. The conventional transformers exhibit a tendency for patch tokens to prioritize their own information, and the presence of residual connections, as mentioned earlier, can lead to trivial solutions of "identical mapping," where the model simply duplicates the input data, neglecting semantic considerations. In contrast, our partial semantic aggregation transformer leverages learnable auxiliary aggregation sequences that effectively capture the normality semantics of the original signal. As depicted in the right-hand side of Fig. 4(c), the aggregation tokens aggregate normality information from the disjoint original patch tokens during the encoding stage, guaranteeing high-level semantic learning through long-range dependencies. As anomalous information is not included in the aggregation process during training, it is effectively filtered out. This mechanism successfully prevents the occurrence of trivial abnormal reconstruction phenomena. This method can be understood as an implicit self-supervised task, wherein the aggregation tokens leverage the global semantic capture ability of the transformer to reconstruct feature patch tokens. This reconstruction is achieved by considering the feature patch tokens that are disjoint from the target position and incorporating the long-range semantics dependencies. Consequently, it offers an efficient reconstruction-based VAD approach that effectively avoids abnormal reconstruction.

E. Objective Function for Training

In the proposed framework, only the feature reconstruction module requires optimization. To accomplish this task, we employ a loss function that considers both the distance and direction within the high-dimensional feature space. Specifically, given the input representation $\mathcal{F} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$ and its corresponding reconstruction obtained through PSA-VT, denoted by $\tilde{\mathcal{F}} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$, the reconstruction error can be measured as follows:

$$\ell(\mathcal{F}, \tilde{\mathcal{F}}) = \frac{1}{\mathcal{H}\times\mathcal{W}} \times \left\{ \left(\left\| \mathcal{F} - \tilde{\mathcal{F}} \right\|_2^2 \right) + \lambda \times \left(1 - \frac{\mathcal{F}\cdot\tilde{\mathcal{F}}}{\|\mathcal{F}\| \times \|\tilde{\mathcal{F}}\|} \right) \right\}. \tag{4}$$

The relative contributions of the two loss terms are balanced by the hyperparameter λ, which is empirically set to 5.

F. Anomaly Detection for Inference

Given that PSA-VT only employs normal samples during the training phase for optimization, its capability lies in accurately reconstructing normal samples. However, abnormal samples that deviate from the established normal semantics tend to induce substantial reconstruction errors. Consequently, the disparity between the extracted representation of an input sample and its corresponding reconstruction can be leveraged as a criterion for anomaly detection

$$\mathcal{A} = \left\| \mathcal{F} - \tilde{\mathcal{F}} \right\|_2^2 \otimes \left(1 - \frac{\mathcal{F}\cdot\tilde{\mathcal{F}}}{\|\mathcal{F}\| \times \|\tilde{\mathcal{F}}\|} \right) \tag{5}$$

where $\mathcal{A} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}}$ is the anomaly score map and the $\otimes$ denotes the element-wise multiplication. For each position, the channel-wise mean value is calculated and assigned as the anomaly score. To match the spatial resolution of the anomaly map with that of the input image, an interpolation operation is employed. Furthermore, to suppress noise, the final anomaly map undergoes a Gaussian filtering process. Subsequently, we choose to utilize the standard deviation of the anomaly score map as the image-level anomaly score since the anomalous samples obviously have high reconstruction errors.

G. Interpretability

In the industrial domain, the interpretability of deep learning models, particularly with regard to black-box models, holds significant importance. This is primarily due to the necessity of model debugging during adaptation to new environments. Blind debugging processes can lead to substantial resource wastage in terms of both manpower and time. Fortunately, our PSA-VT model stands out as a highly interpretable solution.

First and foremost, PSA-VT boasts a minimal number of hyperparameters that require adjustment. When it comes to training, the only hyperparameter in consideration is the loss weight parameter λ, which exerts limited influence on the model's detection performance. Furthermore, PSA-VT's training procedure negates the need for collecting defect samples and eliminates the cumbersome labeling processes that often accompany traditional training workflows. In contrast, the PSA-VT training phase only calls for the collection of a specified quantity of defect-free samples. Most importantly, the interpretability of PSA-VT can be substantiated through the direct visualization of its feature reconstruction results (see Section IV-F4). Diverging from the abstract features generated by conventional classification-based methods, PSA-VT adopts the reconstruction paradigm and effectively addresses the shortcut learning problem in traditional approaches. As a result, the feature reconstruction outcomes provided by PSA-VT offer a direct and comprehensible means of validating the model's training progress, making it a useful tool for practitioners.

IV. Experimental Verification

A. Dataset Details

In this section, we undertake an evaluation of the PSA-VT's performance on various publicly available benchmarks to ascertain its effectiveness. To be precise, we utilized three industrial datasets: Mvtec AD [24], Mvtec 3D [25], and

TABLE I
SUMMARY OF EXPERIMENTAL INDUSTRIAL AND SEMANTIC DATASETS

Dataset / Configuration	Sample Attribute	Industrial Dataset			Sample Attribute	Semantic Dataset		
		Mvtec AD	Mvtec 3D	BTAD		MNIST	F-MNIST	CIFAR-10
Training	Normal number	3629	2656	1799	Total number	60000	60000	50000
	Abnormal number	×	×	×				
Testing	Normal number	467	249	451	Total number	10000	10000	10000
	Abnormal number	1258	948	290				
Category		(15)	(10)	(3)		(10)	(10)	(10)
Image Size	256×256							

beanTech Anomaly Detection (BTAD) [26], as well as three semantic datasets: MNIST [27], Fashion-MNIST [28], and CIFAR-10 [29], for the purpose of this assessment. Table I presents the concise summaries of the datasets utilized in our investigation. In order to ensure a fair comparison, we adhered to the dataset splits as originally outlined in the respective publications, thereby creating dedicated training and testing sets, and no data augmentation was used. For more comprehensive insights into the specifics of these datasets, we encourage referring to the corresponding references.

The Mvtec AD [24] dataset has gained significant popularity in the domain of unsupervised VAD. This dataset encompasses 15 categories, including ten object categories and five texture categories, encompassing a total of 3629 defect-free training images, as well as 1258 defective images featuring various defects and 467 defect-free images for testing. Notably, in order to verify the scalability of different methods in a single-model-multicategory scenario, we adopt a strategy of training all 15 categories simultaneously, deviating from existing methods that train on individual categories separately.

Recently, Mvtec Software GmbH introduced the MVTec3D [25] dataset, which serves as a fresh benchmark primarily designed for the detection of geometric anomalies in 3-D objects. This dataset encompasses both RGB images and 3-D scans. MVTec3D comprises ten distinct classes, all of which pertain to object categories. It is worth noting that in the context of identifying anomalies within MVTec3D, we exclusively leverage RGB images as input data, without incorporating 3-D scans, as our PSA-VT framework operates seamlessly with RGB images.

The experimental evaluation also encompasses the utilization of the BTAD dataset [26], encompassing a comprehensive set of 2830 images categorized across three distinct classes. Analogous to the Mvtec AD dataset, the training subset within each category exclusively comprises nondefective samples, while the test subset encompasses a mixture of nondefective and defective samples. We also assessed the model's performance within a single-model-multicategory scenario in this dataset.

Moreover, the present study extends the task of anomaly detection to include semantic anomaly detection, which necessitates a comprehensive understanding of object semantics. For this purpose, we employed the MNIST [27] dataset, fashion-MNIST [28] dataset, and CIFAR-10 [29] datasets. These datasets are semantically categorized into ten distinct subcategories. We employ a methodology akin to cross validation, wherein, during the training phase, a single subclass is exclusively designated for normal instances, while the remaining classes are considered as abnormal instances and are not utilized for training. The model's performance is subsequently evaluated by calculating the mean across all ten subclasses.

B. Evaluation Criteria

For evaluating the performance of anomaly detection at both the image level and pixel level, we follow the established practice in previous studies. The area under the receiver operating characteristic curve (AUROC) is employed as the primary quantitative evaluation metric, along with the average precision (AP). In addition, we also employed the normalized area under the per-region overlap curve (AUPRO) [30] with a threshold of 0.3 to serve as a more refined indicator of localization accuracy. Higher values for all metrics indicate better performance in anomaly detection.

C. Implementation Setup

In the local feature representation module, we leverage the existing knowledge of a pretrained CNN to extract local descriptors. Specifically, we utilize the EfficientNet-b4 CNN model as our default choice. To achieve multiscale feature extraction, we extract feature maps from the first four convolution modules of the model. These intermediate feature maps are initially resized to a consistent spatial size of 32×32 and subsequently fused in the channel dimension. Consequently, the input image $\mathcal{I}$ is transformed into a high-dimensional tensor $\mathcal{F} \in \mathbb{R}^{32\times 32\times 272}$.

In the global feature reconstruction module, we employed a transformer architecture similar to the masked AE (MAE) [20]. To optimize computational efficiency, we reduced the embedding dimension D to 240. In the context of the partial semantic aggregation paradigm, we conducted embedding of the obtained representations at the patch level, with a patch size of $P = 2$. Furthermore, the default setting for the number of subsets N in the partition is set to 4.

Regarding the architectural implementation of PSA-VT, we have chosen to utilize the PyTorch 1.7.0 deep learning framework. Specifically, for the implementation of the EfficientNet and MAE network structures, we have adopted existing publicly available implementations[1,2] to ensure consistency

[1]https://github.com/facebookresearch/mae
[2]https://github.com/lukemelas/EfficientNet-PyTorch

TABLE II

AUROC AND AP RESULTS OF DIFFERENT METHODS IN MVTEC AD AT BOTH THE IMAGE LEVEL AND PIXEL LEVEL

Category	FastFlow(Arxiv2021)		Draem(ICCV2021)		Padim(ICLR2021)		Patchcore(CVPR2022)		RD4AD(CVPR2022)		**PSA-VT(OURS)**	
	AUC_I/AUC_P	AP_I/AP_P	AUC_I/AUC_P	AP_I/AP_P	AUC_I/AUC_P	AP_I/AP_P	AUC_I/AUC_P	AP_I/AP_P	AUC_I/AUC_P	AP_I/AP_P	AUC_I/AUC_P	AP_I/AP_P
Carpet	82.7 / 95.4	94.1 / 18.6	96.1 / 96.7	98.7 / 69.8	99.6 / 98.3	99.9 / 61.2	99.3 / 98.9	99.7 / **65.3**	96.5 / 98.7	99.0 / 56.2	**100** / **99.1**	**100** / 65.0
Grid	83.3 / 91.8	89.2 / 08.1	97.5 / 96.3	99.1 / **46.6**	70.3 / 74.1	87.0 / 06.7	69.3 / 86.2	86.3 / 12.7	**98.8** / **99.2**	**99.6** / 45.2	98.7 / 95.3	**99.6** / 33.3
Leather	94.6 / 97.3	98.1 / 18.0	98.2 / 97.5	99.4 / **65.9**	99.5 / 99.0	99.8 / 53.9	**100** / 99.1	**100** / 42.4	**100** / **99.4**	**100** / 40.7	**100** / **99.4**	**100** / 47.9
Tile	97.5 / 86.9	99.1 / 26.9	98.9 / 87.6	99.8 / **67.1**	89.8 / 84.5	96.4 / 35.2	98.8 / 94.9	99.6 / 52.5	96.2 / **95.3**	99.1 / 50.1	**100** / 93.2	**100** / 45.4
Wood	99.3 / 87.6	98.1 / 18.0	97.2 / 87.4	99.2 / 46.6	99.3 / 93.5	**99.8** / 40.5	98.1 / 93.4	99.5 / 44.6	**99.4** / **96.2**	**99.8** / **53.1**	98.5 / 94.4	99.6 / 48.2
Bottle	98.9 / 90.0	99.6 / 28.9	87.1 / 76.6	95.5 / 36.9	98.5 / 95.7	99.6 / 64.6	97.3 / 97.2	99.2 / 70.8	99.6 / 97.7	99.9 / 60.9	**100** / **98.5**	**100** / **78.7**
Cable	85.7 / 89.2	90.4 / 22.0	60.4 / 61.8	72.4 / 05.8	71.9 / 82.3	80.9 / 19.4	80.5 / 90.9	89.2 / 30.8	80.2 / 84.0	89.9 / 27.7	**93.9** / **97.1**	**96.1** / **48.6**
Capsule	75.1 / 96.8	93.8 / 22.8	51.9 / 47.9	85.7 / 03.1	78.1 / 91.1	97.4 / 30.4	66.7 /97.7	91.2 / 36.5	**96.5** / 96.3	**99.3** / 41.9	87.7 / **97.9**	96.5 / **44.4**
Hazelnut	97.1 / 93.6	98.3 / 29.3	92.8 / 96.1	96.0 / 41.6	90.6 / 96.9	94.0 / 53.1	97.6 / 97.2	98.9 / 43.1	**100** / **98.9**	**100** / 62.1	99.4 / 98.8	99.6 / **63.6**
Meta nut	90.6 / 92.8	97.7 / 59.7	55.8 / 68.1	88.7 / 21.3	81.4 / 90.1	95.3 / 60.2	91.8 / **97.2**	98.2 / **86.5**	95.9 / 93.7	**99.8** / 62.5	**97.5** / 95.6	99.2 / 61.3
Pill	79.4 / 87.9	95.1 / 24.0	56.5 / 87.2	88.6 / 38.1	68.0 / 91.0	93.1 / 26.8	72.1 / 96.1	93.6 / **73.7**	**94.2** / **96.4**	94.4 / 56.0	91.1 / 92.2	**98.3** / 32.9
Screw	63.4 / 92.8	82.1 / 04.1	73.3 / 93.4	87.2 / 17.1	61.7 / 95.3	83.1 / 05.1	55.3 / 92.9	79.1 / 02.4	**95.8** / **99.4**	**98.5** / **45.3**	92.1 / 98.6	96.7 / 23.6
Toothbruth	86.4 / 96.2	94.8 / 18.0	93.6 / 94.7	97.8 / 33.7	97.2 / 97.4	98.9 / 46.1	79.7 / 98.0	92.2 / 53.3	**99.4** / 97.9	**99.8** / **56.5**	94.7 / **98.9**	97.8 / 50.4
Transistor	88.4 / 89.8	83.1 / 33.3	74.8 / 69.1	75.1 / 18.0	83.5 / 88.5	80.2 / 34.5	95.0 / 92.3	95.6 / 56.1	90.4 / 86.8	92.5 / 35.6	**99.9** / **98.6**	**98.4** / **81.5**
Zipper	84.9 / 90.4	93.7 / 16.2	**99.3** / 96.7	**99.8** / **66.2**	82.9 / 96.1	93.6 / 36.6	95.2 / 97.9	98.8 / 58.2	98.3 / **98.0**	99.7 / 56.1	92.5 / 95.8	97.3 / 38.4
Average	87.1 / 91.9	89.3 / 28.2	82.3 / 83.8	92.2 / 39.3	84.8 / 91.6	93.2 / 38.3	85.5 / 95.4	94.7 / 48.6	96.1 / 95.8	98.1 / 49.9	**96.4 / 96.9**	**98.6 / 50.9**

[1] The best item in the comparison is highlighted in bold, while the second-best item is underlined.
[2] Each result consists of the following four items: image-level AUROC, pixel-level AUROC, image-level AP, and pixel-level AP.

and accuracy. Furthermore, the source code of PSA-VT is openly accessible. Interested readers and researchers can access the code at the following link.[3]

The representation module within the entire model does not require training. However, we train the reconstruction module for a total of 400 epochs using the AdamW optimizer with a batch size of 8 and a learning rate of 1e−4. Each image in the dataset undergoes preprocessing steps, including resizing it to 256 × 256 dimensions and normalization using the mean and standard deviation derived from the ImageNet dataset. All experiments are performed on a desktop computer with an Intel Xeon[4] Gold 6226R CPU operating at 2.90 GHz and two NVIDIA A100 GPUs having 40 GB of memory.

D. Comparative Experiments

In this section, we undertake an extensive comparative analysis of the PSA-VT model in contrast to existing methodologies. To commence, we assess the performance of PSA-VT against SOTA models in the proposed scenario of multicategory scalability training using industrial datasets. Furthermore, we delve into comparative experiments on semantic anomaly datasets to validate the high-level semantic discrimination capability of PSA-VT.

It is crucial to highlight that the performance results reported for the compared methods in their original publications are based on the conventional single-category training paradigm. In order to ensure an equitable evaluation within the context of scalable multicategory training techniques, we rigorously conducted assessments by replicating the compared methods using publicly accessible source code. To elaborate, during the training phase, we trained a comprehensive model on a dataset encompassing all categories within that dataset. It should also be emphasized that no fine-tuning was employed during the testing phase.

[3] https://github.com/hmyao22/PSA

[4] Registered trademark.

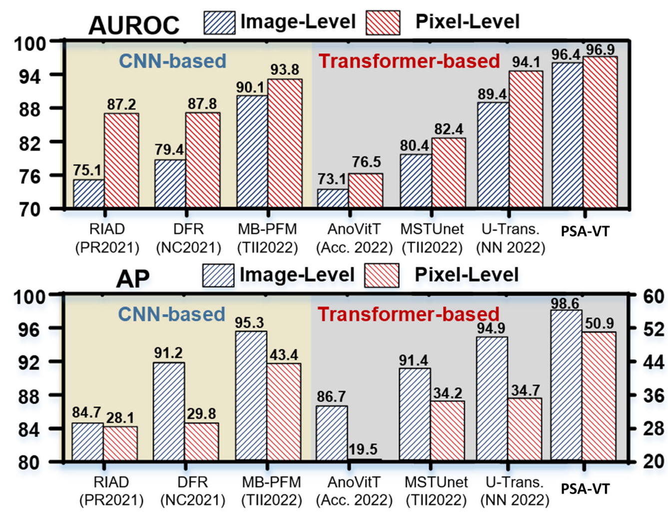

Fig. 5. Comparison results of PSA-VT with the existing advanced reconstruction-based methods on Mvtec AD dataset.

1) Mvtec AD Comparison Result: The PSA-VT model is first compared with several SOTA methods on Mvtec AD dataset under the multicategory scalability training scenario, such as FastFlow [31] which utilizes normalizing flow, the two-stage reconstruction-discrimination method Draem [16] that employs simulated defects, Padim [9] that is based on normal distribution modeling, Patchcore [8] that uses normal sample memory, and the RD4AD [32] method based on inverse knowledge distillation.

Table II presents the quantitative results of the experiments. The proposed PSA-VT model achieves image-/pixel-level AUROC scores of 96.4/96.9 and image-/pixel-level AP scores of 98.6/50.9 across all 15 categories. Compared with the suboptimal RD4AD method, PSA-VT improves the image-/pixel-level AUROC by 0.3 and 1.1 and the image-/pixel-level AP by 0.5 and 1.0, respectively, and significantly outperforms the other competing methods. PSA-VT obtains the highest or second-highest combined accuracy in

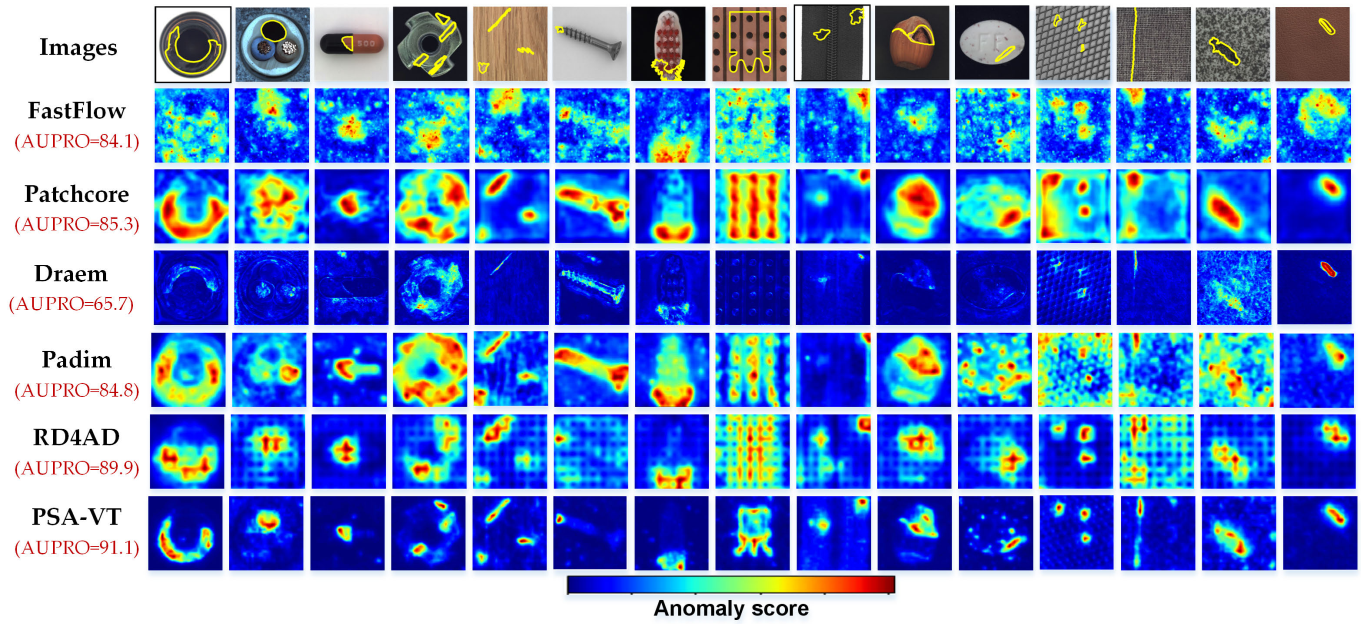

Fig. 6. Qualitative results of various methods for anomaly localization in the scalable running mode on the Mvtec AD dataset. Ground truth annotations are indicated by yellow outlines.

ten categories out of all categories. Notably, PSA-VT shows significant performance improvement for some of the most challenging categories with complex semantics, such as transistor and cable classes.

Since our proposed method belongs to the reconstruction-based category, and to evaluate its performance, we compared it with several SOTA methods within the same category. These methods include RIAD [13], DFR [17], and multihierarchical bidirectional pretrained feature mapping (MB-PFM) [1], which are CNN-based approaches. In addition, we considered AnoViT [33], MSTUet [23], and U-transformer [22], which are transformer-based approaches.

RIAD addresses image reconstruction by formulating it as an inpainting problem, aiming to avoid reconstructing defects in the process. DFR, on the other hand, goes beyond simple image reconstruction and focuses on feature reconstruction.

Utilizing mappings between different pretrained feature spaces, MB-FPM is a recent advanced model. AnoViT employs a transformer encoder to capture global semantics during image reconstruction. In the MSTUet framework, the image inpainting model leverages the long-distance semantic capture capability of the Swin transformer. Abnormal segmentation is accomplished by incorporating synthetic defects and a CNN-based discriminant network. Finally, the U-transformer adopts an U-shaped transformer architecture to facilitate feature reconstruction.

As illustrated in Fig. 5, our method demonstrates superior performance compared with other reconstruction-based approaches. Specifically, compared with the suboptimal MB-PFM [1], we achieve a significant improvement of 6.3/2.6 in AUROC and 3.3/7.5 in AP.

Furthermore, the qualitative detection outcomes of several samples are illustrated in Fig. 6. The results reveal that DRAEM exhibits significant missed detections, while other methods produce imprecise localization outcomes. In contrast, our approach demonstrates the most precise and accurate detection outcomes. We also provide quantitative comparison results of the AUPRO [30] to the false-positive rate of 0.3 for each method, which further highlights the superiority of our proposed scheme in accurately localizing defects.

2) Mvtec 3-D Comparison Result: Table III provides a detailed breakdown of pixel-level AUPRO results, focusing on category-by-category comparisons between PSA-VT and several other methods, including baseline methods such as GAN, AE, and VM [25], as well as the recent SOTA methods 3-D ST [34]/2-D ST [30]. It is noteworthy that while 2-D ST and PSA-VT utilize 2-D RGB information, the other methods rely on 3-D information derived from voxels, depth maps, or point clouds. The results showcase the performance of PSA-VT under individual single-category training [PSA-VT(I)] and scalable multicategory training settings [PSA-VT(S)], highlighting its SOTA capabilities, surpassing both the baselines and the recent SOTA method ST. Some qualitative results are displayed in Fig. 7.

Several key observations can be gleaned from the experimental results. First, despite exclusively utilizing RGB pixel information, PSA-VT outperforms other methods that rely on stereo information, such as the suboptimal method 3-D ST. This finding underscores the effectiveness of using RGB information alone for detecting stereo geometric defects, primarily because geometric anomalies often manifest as alterations in RGB values. While the combination of depth and RGB information from various modalities can enhance localization performance, it is important to note that stereo vision sensors incur additional costs. PSA-VT's ability to achieve superior results solely with RGB pixel information underscores its cost-effectiveness and versatility.

Furthermore, PSA-VT exhibits comparable performance in both the individual single-category training paradigm [PSA-VT (I)] and the scalable multicategory training

TABLE III
QUANTITATIVE COMPARISON OF AUPRO FOR VARIOUS METHODS ON THE MVTEC 3-D DATASET

Modality	Category	Bagel	Cable gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Average
Voxel	GAN	44.0	45.3	82.5	75.5	78.2	37.8	39.2	63.9	77.5	38.9	58.3
	AE	26.0	34.1	58.1	35.1	50.2	23.4	35.1	65.8	01.5	18.5	34.8
	VM	45.3	34.3	52.1	69.7	68.0	28.4	34.9	63.4	61.6	34.6	49.2
Depth	GAN	34.3	52.1	69.7	68.0	28.4	34.9	63.4	61.6	34.6	49.2	14.2
	AE	14.7	06.9	29.3	21.7	20.7	18.1	16.4	06.6	54.5	14.2	20.3
	VM	28.0	37.4	24.3	52.6	48.5	31.4	19.9	38.8	54.3	38.5	37.4
Point Cloud	3D-ST	95.0	48.3	98.6	92.1	90.5	63.2	94.5	98.8	97.6	52.4	83.1
RGB	2D-ST	93.2	91.0	**97.5**	**91.7**	89.8	69.8	93.3	**95.5**	94.5	88.4	90.5
	PSA-VT(I)	94.1	**93.8**	92.3	90.4	94.8	**70.6**	**95.3**	93.2	95.8	**91.6**	**91.2**
	PSA-VT(S)	**94.2**	93.4	91.9	90.5	**94.9**	70.4	95.1	93.4	**96.0**	91.4	91.1

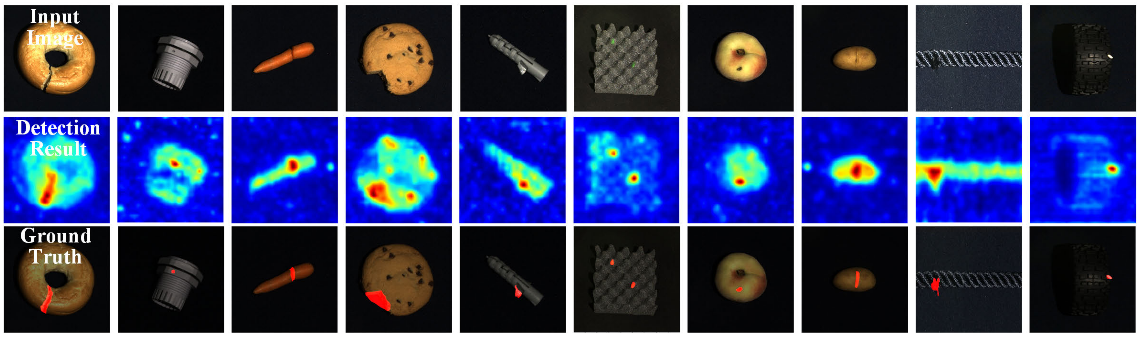

Fig. 7. Qualitative results of PSA-VT for anomaly localization on the Mvtec 3-D dataset.

TABLE IV
COMPARATIVE RESULTS FOR THE BTAD DATASET

Venue-Year	Methods	Image/Pixel AUROC	
		Single-category	Muti-category
ACCV-2020	PatchSVDD	85.2/76.5	80.1/74.6
NC-2021	DFR	95.0/97.1	**94.8**/96.9
ICLR-2021	Padim	94.3/96.2	93.8/96.6
ICCV-2021	Draem	94.4/92.2	91.2/91.9
CVPR-2021	MSFD	91.6/96.2	89.7/96.2
WACV-2022	CFLOW	93.1/97.1	93.0/96.6
AAAI-2023	PMAD	93.6/97.4	93.8/97.3
	PSA-VT(ours)	**95.5/97.6**	**95.3/97.6**

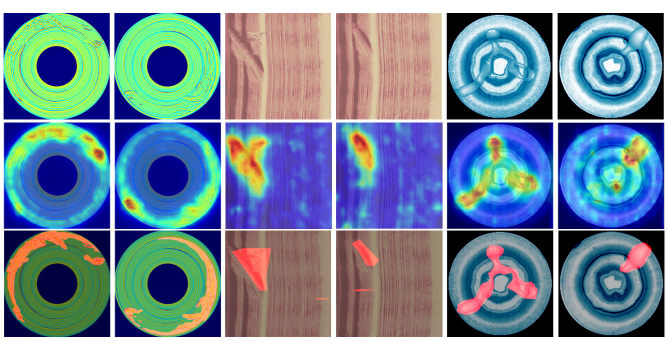

Fig. 8. Qualitative anomaly localization results of PSA-VT on BTAD dataset. From top to bottom: abnormal samples, anomaly score maps, and ground truths.

TABLE V
COMPARATIVE RESULTS FOR THE HIGH-LEVEL SEMANTIC ANOMALY DETECTION TASK

Methods	MNIST	F-MNIST	CIFAR-10	Average
Padim(ICLR2021)	67.6	66.7	52.0	62.1
GANomaly(ACCV2018)	92.8	80.9	69.5	81.1
LSA(CVPR2019)	97.5	92.2	64.1	84.6
ARAE(NN2021)	97.5	93.6	71.7	87.6
ARNet(TMM2020)	**98.3**	93.9	**86.6**	**92.9**
PSA-VT(ours)	97.1	**94.1**	74.7	88.6

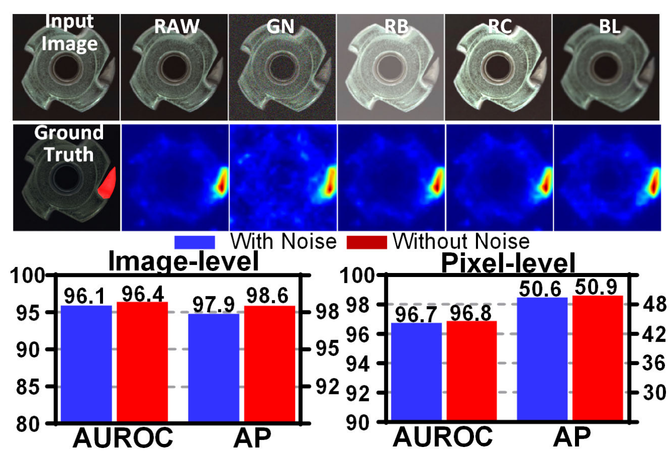

Fig. 9. Antinoise robustness test results of PSA-VT. Qualitative detection results of PSA-VT for various noise samples (top). RAW: original image, GN: Gaussian noise, RB: random brightness, RC: random contrast, and BL: blur. Quantitative results of PSA-VT on the Mvtec AD dataset with/without noise (bottom).

paradigm [PSA-VT (S)]. This highlights its robust generalization capabilities, as it accurately detects product anomalies across all ten categories using a common model weight.

3) BTAD Comparison Result: Table IV lists a qualitative comparison of PSA-VT with several existing SOTA methods PatchSVDD [35], DFR [17], Padim [9], Draem [16], MSFD [36], CFLOW [37], and PMAD [38] in terms of anomaly detection and localization for the BTAD dataset. We present

TABLE VI
QUANTITATIVE COMPARISON RESULTS OF MODEL ROBUSTNESS IN THE SCALABLE MODE, RUNNING EFFICIENCY, AND MEMORY CONSUMPTION

Category/Methods			Generalizable robustness(Image-/pixel-level AUROC)			Efficiency and memory consumption		
			Individual	Scalable	Reduction	Parameters	FLOPs	FPS
SOTA Methods		FastFlow	**99.4/98.5**	87.1/91.9	-12.3%/-06.7%	66.17 Mb	36.22 Gb	**54.2**
		Draem	98.0/97.3	82.3/83.8	-16.0%/-13.8%	97.42 Mb	198.66 Gb	52.8
		Padim	95.3/97.5	84.8/91.6	-11.0%/-06.1%	68.88 Mb	366.58 Gb	4.6
		Patchcore	99.2/98.4	85.5/95.4	-13.8%/-03.0%	68.88 Mb	11.43 Gb	38.9
		RD4AD	98.5/97.8	96.1/95.8	-02.4%/-02.0%	91.75 Mb	31.61 Gb	37.2
Reconstruction Methods	CNN Model	RIAD	91.7/94.2	75.1/87.2	-18.1%/-07.4%	28.81 Mb	694.74 Gb	10.3
		DFR	93.8/95.5	84.7/87.8	-09.7%/-08.1%	33.93 Mb	82.45 Gb	50.2
		MB-PFM	97.5/97.3	90.1/93.8	-07.6%/-03.6%	68.91 Mb	22.90 Gb	19.3
	Transformer Model	AnoViT	78.0/83.0	73.1/76.5	-06.3%/-07.8%	93.87 Mb	65.49 Gb	34.6
		U-Transformer	96.0/96.7	89.4/94.1	-06.9%/-02.6%	63.44 Mb	82.56 Gb	10.7
		MSTUnet	98.9/96.4	80.4/82.4	-18.7%/-14.5%	55.49 Mb	61.49 Gb	5.3
		PSA-VT(ours)	96.6/97.0	**96.4/96.9**	**-00.2%/-00.1%**	**14.59 Mb**	**10.17** Gb	13.7

results for both single-class training and multiclass training scenarios. In both cases, PSA-VT demonstrates the optimal performance in image-level anomaly detection and pixel-level anomaly localization. In comparison to the most recent PMAD (AAAI2023) method, PSA-VT exhibits notable improvements in image pixel-level AURPO. Specifically, we achieve a gain of +1.9/0.2 and +1.5/0.3 in the two respective training modes, underscoring the effectiveness of our approach. Fig. 8 provides illustrative examples of anomaly localization performed by PSA-VT on the BTAD dataset.

4) Semantic Anomaly Comparison Result: Our PSA-VT approach introduces auxiliary aggregation tokens to transform the low-level reconstruction task into a high-level regression task, which is particularly significant for semantic-level anomaly detection. In this section, we present an evaluation of the performance of PSA-VT on semantic anomaly detection. To this end, we compare the results obtained by PSA-VT with those obtained by some SOTA methods for semantic anomaly detection, such as GANomaly [39], LSA [40], ARAE [41], and ARNet [42], as well as Padim [9], a method designed for industrial defect detection.

The results of the comparison are presented in Table V. It is observed that while Padim shows promising outcomes in low-level tasks such as industrial anomaly detection, it suffers from significant performance degradation in high-level semantic anomaly detection tasks. This suggests that Padim is only effective in detecting anomalies with local structural damage and is unable to handle semantic anomalies. Compared with the other methods for semantic anomaly detection, our approach ranks second, with suboptimal performance compared with ARNet. However, according to the original research paper of ARNet [42], the method only achieved 83.9 AUROC for industrial anomaly detection tasks, which is much lower than the outcomes achieved in this study.

E. Model Robustness and Efficiency Assessment

In this section, we aim to assess the robustness and efficiency of the model on several fronts. First, we will evaluate the model's robustness against noise. Second, we will explore the model's scalability robustness, particularly when transitioning from single-class anomaly detection to the multiclass anomaly detection scenario as proposed. Finally, we will conduct an assessment of the computational complexity of the model.

1) Robustness Evaluation Against Noise: In this experimental investigation, we examined the influence of noise introduction on the data on the performance of PSA-VT. Specifically, we applied random data augmentation techniques using the Albumentations library [43] to process image samples. These augmentation techniques included the introduction of Gaussian noise, random alterations in brightness and contrast, as well as blurring. The resulting images with added noise were then used for testing purposes.

The top section of Fig. 9 visually illustrates the qualitative impact of several typical noise types on the detection outcomes achieved by the PSA-VT method, while the bottom section provides quantitative metrics for evaluation. Our findings indicate that the presence of noise in the sample images does not significantly compromise the overall performance of PSA-VT. This observation underscores the robustness of our approach to handling noisy data.

2) Robustness Evaluation of Model Scalability: In order to address the challenges faced by traditional VAD methods in real industrial scenarios, we propose a novel approach in this study: the scalable operating mode. Traditional VAD methods often encounter memory issues as they require storing model parameters separately for each category. In contrast, the scalable VAD method only requires a single set of parameters for different types, making it more suitable for high-efficiency requirements in industrial environments.

However, the scalable operating mode places high demands on the robustness of model scalability, as it requires the model to learn normal distributions of multiple types simultaneously, which presents a greater challenge. In order to thoroughly evaluate the models, we conducted evaluations under two distinct modes of operation: the "Individual" mode, which aligns with the traditional approach of employing one model per category, and the "Scalable" mode, which represents a novel strategy of utilizing a single model for multiple categories.

The experimental results, as presented in Table VI, indicate the performance of our proposed PSA-VT method under both the traditional "Individual" mode and the more challenging "Scalable" mode. In the traditional "Individual" mode, PSA-VT achieves competitive results compared with SOTA models, such as FastFlow, Draem, and Patchcore, although it does not surpass them. However, when the existing methods are switched to the "Scalable" mode, significant performance drops are observed in both image-level and pixel-level detection accuracy for the compared models. For instance, the SOTA model FastFlow experiences a drop of 12.3 and 6.7, and Patchcore experiences a drop of 13.8 and 3.0. This performance degradation can be attributed to the fact that these methods primarily focus on extracting local information while neglecting global semantics. The performance degradation of RD4AD, which incorporates a bottleneck structure to capture global semantics, is mitigated to some extent.

In contrast, our PSA-VT method exhibits minimal performance degradation when transitioning to the more demanding "Scalable" mode. This impressive attribute can be attributed to the effective capture of high-level global semantics by our method, underscoring its robust scalability and superior proficiency in addressing complex VAD challenges across multiple categories. Exploring an even more rigorous scenario, we conducted training on all categories encompassed by the three datasets (Mvtec AD, Mvtec 3-D, and BTAD) under examination utilizing a single model. Notably, we observed that the model's detection accuracy remained relatively unaffected, providing further evidence of the robust generalization capabilities of our approach.

3) Efficiency and Consumption Evaluation: In addition to performance evaluation, assessing model efficiency and memory consumption are crucial. In this regard, we will conduct evaluations from the above two perspectives. For efficiency, we will utilize floating point operations per second (FLOPs) and frames/s (FPS) as metrics to quantify computational efficiency. As for memory consumption, we will employ the number of parameters as a metric to gauge model complexity.

The comparison results are presented in Table VI, showcasing the efficiency and memory consumption analysis. In terms of efficiency, it is observed that PSA-VT, utilizing the transformer model, exhibits a lower inference speed compared with CNN-based methods, such as FastFlow and Dream. However, the model still achieves a commendable 13.7 FPS, which meets the requirements for real-world industrial deployment. Concerning memory consumption, PSA-VT demonstrates a lightweight architecture, employing merely 14.59M parameters and 10.17G FLOPs. This makes it the most resource-efficient model among the compared methods. While other competing approaches employ more complex model structures, their performance experiences a more substantial decline when encountered with above mentioned scalable running mode. Notably, despite its lightweight design, PSA-VT maintains high levels of generalization robustness, due to its capacity to extract global high-level semantics.

F. Ablation Studies

1) Influence of the Feature Representation: In this framework, we leverage the inherent local semantic capture ability of CNNs and introduce a local feature representation module aimed at converting image pixels into feature representations.

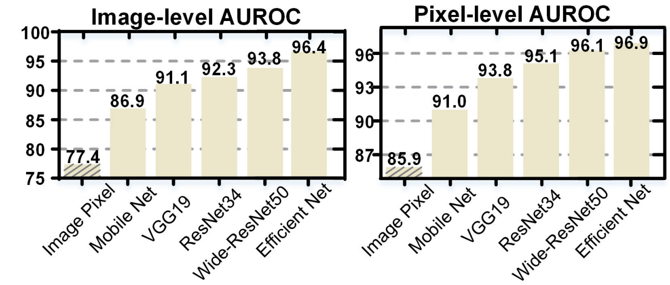

Fig. 10. Impact of feature representation on image-/pixel-level detection performance.

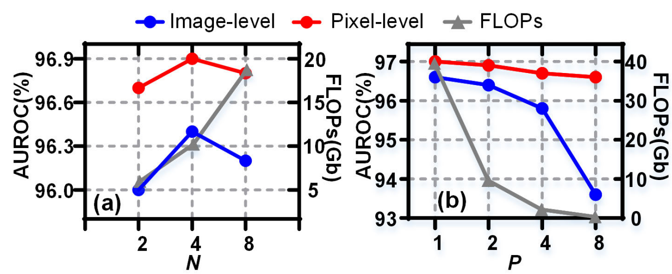

Fig. 11. Ablation experiments of the effect of hyperparameter subset number N and patch size P on model performance and computational efficiency. (a) Effect of N. (b) Impact of P.

In this section, we shall examine the efficacy of this approach by conducting a comparative analysis between image pixels and feature tensors. Furthermore, the performance variations across different types of CNN architectures will also be explored.

The findings are illustrated in Fig. 10, which provides insights into the comparative performance of the reconstruction methods based on feature representations as opposed to image pixels. Notably, the feature-based reconstruction methods consistently outperform the pixel-based counterparts, primarily due to their ability to capture more discriminative local semantics while retaining fewer details. These characteristics enhance the discriminability of defects and reduce the complexity associated with the reconstruction process. Furthermore, the study reveals that different CNN architectures exhibit varying capabilities in representing local semantics. Shallow network architectures tend to demonstrate limited representation abilities, while deeper networks often exhibit a bias toward classification tasks. Notably, when considering the PSA-VT method, EfficientNet-b4 emerges as the optimal choice, primarily due to its exceptional capability in identifying structural damages within defects.

2) Sensitivity of the Subset Number: As outlined in Section III-D, the implementation of a partial semantic aggregation mechanism necessitates the separation of the auxiliary aggregation sequence $\mathcal{X}^*$ into N distinct subsets. Each subset is then aggregated with the semantics of nonintersecting feature patch tokens. The choice of N has a direct impact on both the detection performance and efficiency of the mechanism.

The findings from the ablation experiments regarding the N value are presented in Fig. 11(a). In terms of detection performance, both the image-level and pixel-level AUROC exhibit a trend of initially increasing and then decreasing with the variation of N. This observation can be attributed to the fact that when N is too small, there is a reduced proportion of original feature patch tokens in (3), resulting in insufficient availability of original feature information. Conversely, when N is excessively large, the proportion of aggregation tokens in (3) decreases, impeding the complete aggregation of feature patch token information. Regarding computational efficiency, it shows an increasing trend as N increases. Consequently, a default value of $N = 4$ has been selected based on these observations.

3) Effect of the Patch Size: In contrast to existing approaches [7], which typically employ direct element-by-element embedding projection on the CNN feature map, our proposed method adopts a patch-level embedding strategy on the feature map to cater to the more computationally efficient requirements in industrial scenarios. In this section, we investigate the impact of this strategy on the detection performance and efficiency.

The experimental results, depicted in Fig. 11(b), demonstrate the effects of varying the value of P, which represents the level of fine grainedness in the embedding process. As P increases, it indicates a gradual decrease in the granularity of the embedding, resulting in a reduction in the length of the generated embedding sequence. Consequently, both the detection accuracy and computational complexity exhibit a corresponding decrease. Generally, a finer-grained embedding method has the potential to enhance accuracy, but at the cost of increased computational complexity.

Observations reveal that when $P = 2$, the detection accuracy is slightly lower compared to $P = 1$, while offering a more noticeable computational complexity advantage. As a result, we have selected a default setting of $P = 2$ for our approach.

4) Effect of the Feature Reconstruction: The primary contribution of this study lies in the introduction of a novel partial semantic aggregation mechanism, which aims to enhance the feature reconstruction process and address the inherent challenge of trivial anomaly reconstruction in existing reconstruction-based VAD methods. To assess the efficacy of this mechanism, we conducted a series of experiments as a demonstration. In particular, we utilized the vanilla transformer model without the partial semantic aggregation paradigm as the baseline model and then evaluated the enhancements achieved by incorporating the partial semantic aggregation paradigm.

Initially, we will visualize the outcomes of the refactoring process. Since PSA-VT employs a feature reconstruction strategy, direct observation is not straightforward. To provide a more intuitive representation, we introduce a convolutional decoder tasked with transforming the reconstructed features into pixel values. The results are depicted in the top section of Fig. 12. It can be observed that in the case of the Mvtec AD dataset, the baseline model utilizes the "identical mapping" of the duplicated input, leading to the reconstruction of abnormal patterns. This is not conducive to the reconstruction-based VAD method. In contrast, our proposed method successfully restores abnormal defects to their normal patterns. This is evident even in challenging scenarios, such as samples with missing transistors, where our method is capable of generating normal patterns.

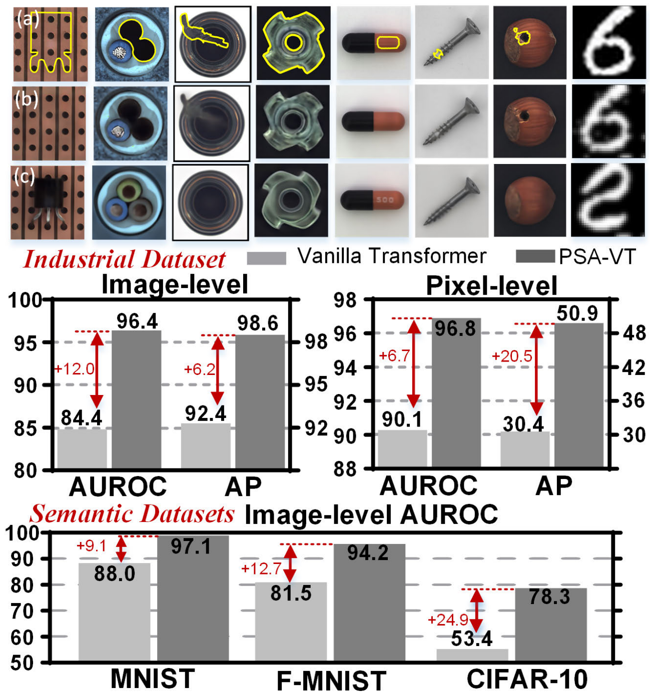

Fig. 12. Effect of feature reconstruction. (a) Original defective samples, (b) reconstruction results using the vanilla transformer, and (c) reconstruction results using PSA-VT (top). The models are trained on digit "2" for the MNIST sample. Quantitative results comparing the vanilla transformer and PSA-VT (bottom).

The quantitative performance improvements resulting from this enhancement are depicted in the top part of Fig. 12. The introduction of the partial semantic aggregation mechanism has yielded significant enhancements to the baseline model. Specifically, the industrial VAD task on the Mvtec AD dataset, it has led to improvements of 12.0 and 6.8 in image-level/pixel-level AUROC detection performance, as well as 6.2 and 20.5 in image-level and pixel-level AP detection performance, respectively, when compared with the baseline. In the case of high-level semantic VAD tasks, the incorporation of the partial semantic aggregation paradigm has resulted in an increase of nearly or exceeding 10 in AUROC detection performance at the image level across the three datasets: MNIST, fashion MNIST, and CIFAR. These findings effectively validate the efficacy of the partial semantic aggregation mechanism in enhancing the reconstruction process.

5) Attention Visualization: This section presents a detailed analysis of the PSA-VT model's attention matrix through visualization. Fig. 13 shows the comparison of the attention matrix of the PSA-VT model with that of a vanilla transformer. The comparison reveals two essential characteristics of the PSA-VT attention matrix. First, the attention matrix of the vanilla transformer shows a strong inclination toward a position-based attention pattern [44], as evidenced by its large diagonal elements. This pattern implies that query tokens primarily focus on themselves and their immediate surroundings. Conversely, the PSA-VT attention matrix has relatively small diagonal elements, indicating that each token

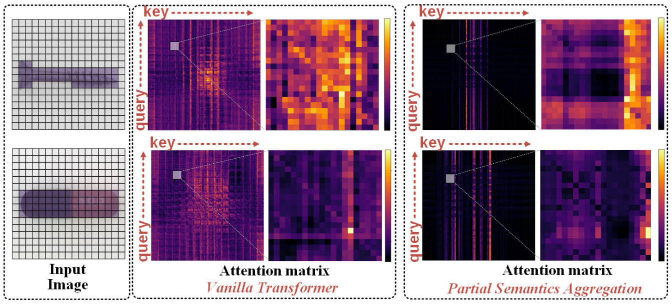

Fig. 13. Attention matrix visualization results. The vanilla transformer model has a higher tendency toward the relative position-based attention pattern [44], while the PSA-VT model displays a stronger inclination toward the content-based attention pattern [44]. This suggests that PSA-VT has a better capability to extract high-level semantics related to objects. Partial enlargements are provided for ease of observation. The attention matrix is obtained by averaging all layers and heads of the encoder.

Fig. 14. AOI equipment for bottle cap defect inspection (left). (a) Overall equipment. (b) Inspection module. (c) Delivery module. Samples of normal and defective bottle caps (right).

has a wider semantic reach and can establish long-range semantic connections with other tokens. This feature allows our method to avoid trivial reconstruction shortcuts caused by a narrow self-attention. Second, the attention matrix displays a content-based attention pattern resembling vertical bars [44], implying that certain key tokens have high attention values for all query tokens. This property suggests that PSA-VT can extract high-level semantics for objects.

G. Practical Application

Fig. 14 presents the implementation of PSA-VT in our automatic optical inspection (AOI) equipment for bottle cap anomaly detection (BC-AD), which further validates the effectiveness of our model.

1) Fast Adaptation With Incremental Learning: It is noteworthy that the generalized ability of the proposed model, a crucial attribute for practical industrial applications, has been demonstrated in this application. Specifically, the proposed model leverages the model trained on the Mvtec AD dataset as the base model and adopts incremental learning [45] for fine tuning when a new type of sample is introduced for inspection. This approach not only saves time and computational resources but also demonstrates the versatility and adaptability of the proposed model in addressing real industrial scenarios.

Drawing inspiration from [45], we have adopted an incremental learning approach based on knowledge distillation.

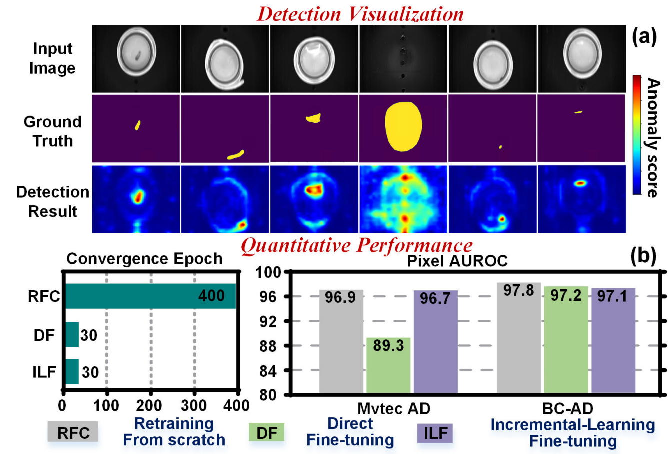

Fig. 15. (a) Detection results of different types of defects by the PSA-VT. (b) Quantitative comparison of the different training approaches.

To elaborate, we define two models—old and new—using the model trained on the Mvtec AD dataset as the foundation. Subsequently, the new model is trained on the BC-AD dataset with a new reconstruction task, while mimicking the reconstruction behavior of the old model on the Mvtec AD dataset to prevent catastrophic forgetting. As a result, the new model is fine tuned using the distillation loss and reconstruction loss

$$\mathcal{L} = \underbrace{\partial \cdot \ell\left(\mathcal{F}_{\mathcal{N}}^{\tilde{\mathcal{M}}}, \mathcal{F}_{\mathcal{O}}^{\tilde{\mathcal{M}}}\right)}_{\text{Distillation}} + \underbrace{(1-\partial) \cdot \ell\left(\mathcal{F}_{\mathcal{N}}^{\tilde{\mathcal{B}}}, \mathcal{F}_{\mathcal{N}}^{\mathcal{B}}\right)}_{\text{Reconstruction}} \tag{6}$$

where $\mathcal{F}_{\mathcal{N}}^{\tilde{\mathcal{M}}}$ and $\mathcal{F}_{\mathcal{O}}^{\tilde{\mathcal{M}}}$ represent the reconstructed features of the old and new models for random exemplars in the Mvtec AD dataset, while $\mathcal{F}_{\mathcal{N}}^{\tilde{\mathcal{B}}}$, $\mathcal{F}_{\mathcal{N}}^{\mathcal{B}}$ denote the extracted and reconstructed features of the new model for new samples in the BC-AD dataset. The weight hyperparameter is represented as ∂.

Fig. 15(a) depicts the inspection outcomes that demonstrate the high accuracy of the proposed approach in detecting all types of BS-AD defects. Fig. 15(b) presents the quantitative performance comparison among different training methods. The results indicate that fine-tuning strategies exhibit a faster convergence rate compared to the retraining method. Furthermore, direct fine tuning on the BC-AD dataset would cause catastrophic forgetting of previously learned knowledge from the Mvtec AD dataset. However, the incremental learning method can prevent this issue while ensuring performance on both datasets.

2) High-Throughput Inference With Efficient Deployment: In real industrial applications, with the aim of harnessing the full potential of inference efficiency, we undertook additional optimization of the PSA-VT model utilizing the TensorRT [46] SDK. As a result, the model's execution speed significantly improved, achieving a commendable rate of approximately 30 FPS. This level of performance aptly satisfies the requirements for high-speed detection within industrial settings.

V. Conclusion and Discussion

In this study, we propose a novel hybrid framework called PSA-VT for scalable multicategory industrial VAD tasks.

PSA-VT consists of two key modules: a local feature representation module based on CNN and a global feature reconstruction module based on ViT. The primary contribution of this article lies in the introduction of a partial semantic aggregation paradigm within the reconstruction module, which significantly enhances the global semantic acquisition capability during the reconstruction process and addresses the limitations of conventional reconstruction-based methods that often suffer from the identical mapping problem.

We have conducted comprehensive evaluations of PSA-VT on multiple widely used benchmarks, including industrial and semantic VAD. The results demonstrate that PSA-VT achieves superior performance across various metrics, particularly in challenging scenarios involving scalable multicategory detection tasks. Furthermore, we have conducted thorough ablation experiments to shed light on the inner workings of the proposed framework. Moreover, we have successfully demonstrated the practical applicability of PSA-VT by achieving rapid adaptation to real-world deployments through incremental learning in real industrial settings.

While the partial semantic aggregation process, being a multisubset parallel operation, does have a certain impact on computational efficiency, we recognize the importance of addressing this issue and mitigating it using deployment tools. In future research, we aim to develop more efficient solutions by improving the intrinsic attention mechanism of the ViT, thereby further enhancing the overall efficiency of the framework.

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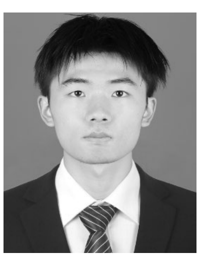
Haiming Yao (Graduate Student Member, IEEE) received the B.S. degree (Hons.) from the School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2022. He is currently pursuing the Ph.D. degree with the Department of Precision Instrument, Tsinghua University, Beijing, China.

His research interests include visual anomaly detection, deep learning, visual understanding, and artificial intelligence for science.

Mr. Yao serves as a reviewer for several journals, such as IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS and IEEE TRANSACTIONS ON IMAGE PROCESSING.



Wei Luo (Student Member, IEEE) received the B.S. degree from the School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2023. He is currently pursuing the Ph.D. degree with the Department of Precision Instrument, Tsinghua University, Beijing, China.

His research interests include deep learning, anomaly detection, and machine vision.

Jianan Lou (Graduate Student Member, IEEE) received the B.Eng. degree in navigation engineering and the B.Ec. degree in economics from Wuhan University, Wuhan, China, in 2022. He is currently pursuing the Ph.D. degree with the Department of Precision Instrument, Tsinghua University, Beijing, China.

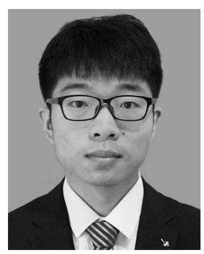
His research mainly focuses on multisensor fusion.

Wenyong Yu (Senior Member, IEEE) received the M.S. and Ph.D. degrees from the Huazhong University of Science and Technology, Wuhan, China, in 1999 and 2004, respectively.

He is currently an Associate Professor with the School of Mechanical Science and Engineering, Huazhong University of Science and Technology. His research interests include machine vision, intelligent control, and image processing.

Xiaotian Zhang received the B.S. degree from Beihang University, Beijing, China, in 2019. He is currently pursuing the Ph.D. degree with the Department of Precision Instrument, Tsinghua University, Beijing.

His main research interests include anomaly detection, generative adversarial networks, and edge computing.



Zhenfeng Qiang received the B.S. degree in mechanical engineering from the Shaanxi University of Science and Technology, Xi'an, China, in 2017, and the M.S. degree in mechanical engineering from Jilin University (JLU), Changchun, China, in 2020. He is currently pursuing the Ph.D. degree with the Department of Precision Instrument, Tsinghua University, Beijing, China.

His research interests include system development of non-dispersive infrared (NIDR) sensors, artificial intelligence, and biomechanics.

Hui Shi (Member, IEEE) received the master's degree from the Huazhong University of Science and Technology, Wuhan, China, in 2002. She is currently pursuing the Ph.D. degree with the School of Mechanical and Electronic Engineering, Wuhan University of Technology, Wuhan.

Her research interests include machine vision, machine learning, and intelligent systems.